

ASAD ALI MIR

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SUMMARY

Computer Science graduate with experience in software development, specializing in Artificial Intelligence and Machine Learning. Experienced in developing AI-driven applications, web-based systems and intelligent automation solutions using modern LLM technologies, with projects involving Retrieval-Augmented Generation (RAG) systems, AI-powered platforms, and full-stack software solutions. Interested in applying AI and software development skills to build practical solutions for real-world challenges.

EXPERIENCE

FriendsBit - Web Development Trainee

Aug 2025 - Oct 2025

- Developed responsive React.js applications using component-based architecture and reusable UI components.
- Applied state management concepts to build maintainable and interactive web interfaces, while gaining exposure to Node.js and backend development concepts.

HiSkyTech - Python Development Intern

July 2024 - Aug 2024

- Developed Python-based projects including a web scraper, rule-based chatbot, and machine learning models such as linear regression and spam classification.
- Performed data preprocessing, cleaning, and transformation using Pandas, NumPy, and BeautifulSoup, and applied basic model evaluation techniques using Scikit-learn and NLTK.

KEY PROJECTS

Assess.io — AI-Powered Multi-Modal Assignment Grading System (FYP)

Oct 2025 - May 2026

- Developed an AI-based platform that automates the grading of source code, PDFs, DOCX files, and handwritten assignment submissions in a unified system.
- Integrated Groq LLaMA 3.3 with custom prompts to generate rubric-based scores, feedback, and readability analysis for student submissions.
- Implemented asynchronous submission processing via Celery and Redis, and a secure Docker-based sandboxing for isolated code execution.
- Integrated FastAPI backend services with secure login, roles-based authorization, and PostgreSQL database to manage courses, assignments, and grades.
- Built a React-based web dashboard for students and instructors to track submission and review AI-generated feedback in real time.

PaperMind - Multi-document Q&A System using RAG

April 2026

- Developed a RAG system using LangChain, ChromaDB, and Llama 3.1, enabling researchers to instantly query and extract insights from multiple academic papers through a conversational interface.
- Built a semantic search pipeline with embeddings and custom prompting to generate accurate, context-aware answers with source attribution.

AI-Driven Urban Traffic Optimization System

May 2025 - June 2025

- Collaborated on a multi-stage AI pipeline combining YOLOv8 vehicle detection, LSTM-based congestion forecasting, VAE-based anomaly detection, and reinforcement learning for traffic signal control.
- Implemented a KeyBERT-based NLP module to detect real-world traffic incidents from social media, alongside fairness and environmental impact auditing to evaluate the resulting traffic control decisions.

SKILLS

Programming: Python, JavaScript, C++

AI/ML: Machine Learning, Deep Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Computer Vision, NLP, Reinforcement Learning

Libraries: Pandas, NumPy, Scikit-learn, OpenCV, NLTK, Sentence Transformers

Frameworks: Django, React.js, Node.js, Langchain, Pytorch, TensorFlow

Tools & Platforms: Git, GitHub, VS Code, Streamlit, Jupyter Notebook, SQL Databases, Vector Databases

Soft Skills: Problem Solving, Analytical Thinking, Communication, Teamwork, Continuous Learning

EDUCATION

University of Management and Technology

2022 - 2026

Bachelor of Computer Science - CGPA: 3.44 / 4.00

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Software Engineering, Database Systems, Digital Image Processing, Operating Systems

CERTIFICATIONS

[Andrew Ng's Machine Learning Specialization \(Coursera\)](#)